

Crossfade: The Mathematics in Three Pages

Jason

2026-09-21 (condensed from *The Mathematics to Check*, spec v0.3)

Every quantity Crossfade estimates is a functional of a named channel representation; regimes are dimensionless coordinates; transfer is a measured effect. Tags: **[G]** the software enforces it, **[P]** a swappable convention, **[H]** a claim the benchmark measures. Nine *Checks* are collected at the end; Checks 3 and 6 are on the critical path. Notation: x, y baseband input and output; $h(t, \tau)$ Bello kernel; f_c, B, T carrier, bandwidth, observation; c medium speed; v closing speed; $M = v/c$; $\beta_B = B/f_c$; arrival list $\{(\tau_k, a_k)\}$.

1 The channel and when each representation is sparse

$$y(t) = \int h(t, \tau) x(t - \tau) d\tau = \iint S_H(\tau, \nu) x(t - \tau) e^{j2\pi\nu t} d\tau d\nu = \iint h_W(\tau, \alpha) \sqrt{\alpha} x(\alpha(t - \tau)) d\tau d\alpha. \quad (1)$$

$S_H = \mathcal{F}_{t \rightarrow \nu} h$ is the delay–Doppler spreading function and exists for any LTV kernel [2, 11]; h_W is the delay–scale form [1], with $\alpha = 1 + v/c$ one-way or $(c + v)/(c - v)$ monostatic **[P]**. Neither is “the” narrowband or wideband form; what depends on regime is sparsity. One path with scale α_0 has drifting delay $\tau(t) = \alpha_0 \tau_0 + (1 - \alpha_0)t$, so at resolution $(1/B, 1/T)$ it occupies about $\max(1, \Pi_s)$ cells along each delay–Doppler axis, where

$$\boxed{\Pi_s = TB |\alpha_0 - 1| = TB M} \quad (\text{range-walk number; Kelly–Wishner } TB 2v/c \ll 1 \text{ [9]}), \quad (2)$$

and is a single point in (τ, α) . A pack declares the threshold on Π_s at which its canonical kind switches from **delay_doppler** to **delay_scale**; it is not core logic. **Realization vs. statistic [G]**: under WSSUS, $\mathbb{E}[S_H(\tau, \nu) S_H^*(\tau', \nu')] = C_H(\tau, \nu) \delta(\tau - \tau') \delta(\nu - \nu')$; the power–delay profile is $P(\tau) = \int C_H d\nu$. Every anchor record says which it holds (**realization, expectation, estimate, posterior_samples, pseudo_label**); WSSUS is a declared parameter, never a default.

2 Descriptors as functionals

For a profile $P \geq 0$: $\bar{\tau} = \int \tau P / \int P$, $\sigma_\tau^2 = \int (\tau - \bar{\tau})^2 P / \int P$; discrete with $w_k = E_k / \sum_j E_j$: $\sigma_\tau^2 = \sum_k w_k (\tau_k - \bar{\tau})^2$. Resolvable-arrival count $N_{\text{arr}}(\Delta, \gamma)$ **[P]**: merge arrivals within Δ greedily in delay order, count clusters with energy $\geq \gamma \max_i E_i$; the rule is recorded and within-cell arrivals are listed as **ambiguous**. For an arrival list, $P(\tau) = \sum_k |a_k|^2 \delta(\tau - \tau_k)$ and both descriptors are exact, which is why W1’s simulator emits the list.

$$\sigma_\tau^2 = \frac{1}{2} \sum_{k,l} w_k w_l (\tau_k - \tau_l)^2, \quad (3)$$

so σ_τ is shift-invariant and determined by the delay-difference profile $Q(\tau) = \sum_{k,l} |a_k|^2 |a_l|^2 \delta(\tau - \tau_k + \tau_l)$. A passive receiver’s autocorrelation $R_{yy} = \sum_{k,l} a_k a_l^* R_{ss}(\tau - \tau_k + \tau_l)$ has exactly those weights, so it recovers σ_τ without the source when pairwise differences exceed $1/B$, and is biased when they collide; W1 measures that bias against exact truth. **[G]**: one string per quantity, **canonical_kind** (or **not_channel**), is the whole mechanism making a sonar descriptor and an HF descriptor comparable.

3 Regime coordinates and the overlap result

$M = v/c$	$\beta_B = B/f_c$	$\Pi_\tau = \sigma_\tau B$	$\Pi_\nu = \sigma_\nu T$
$\Pi_L = L/\lambda$	$\Pi_f = f/f_{\text{cut}}$ (mode count)	$\Pi_h = k h_{\text{rms}} \sin \theta$	SNR_{cell}
sonar; HF placeholders:	$M: 10^{-3}\text{--}10^{-2}; 10^{-6}\text{--}10^{-5}$	$\beta_B: 0.3\text{--}1; 10^{-3}\text{--}10^{-2}$	$\Pi_\tau: 10\text{--}10^3; 1\text{--}50$
	$\Pi_\nu: 0.1\text{--}10; 0.1\text{--}10$	$\Pi_L: 10\text{--}100; 10\text{--}300$	$\Pi_f: 1\text{--}50; 1\text{--}10$

The window question has one answer in both domains: window length follows Π_ν , resolution follows Π_τ . Each descriptor q names a governing set G_q ; tested support $C_d(q)$ is a set of cells in $\log_{10}$ coordinates; overlap $O_q = C_{\text{src}} \cap C_{\text{tgt}}$ and distance $d_q(c) = \min_{c'} \max_{g \in G_q} |\log_{10} \pi_g(c) - \log_{10} \pi_g(c')|$ in decades **[P]**. Support is the list of evaluated cells, never a bounding box **[G]**.

Proposition 3.1 (Overlap result). *Under the placeholders, sonar and HF are two decades apart in M and 1.5 in β_B . For any descriptor with M or β_B in G_q , $O_q = \emptyset$. **[H]**: Doppler and scale descriptors are therefore predicted not to transfer and serve as the negative control; what can transfer lives in $\{\Pi_\tau, \Pi_\nu, \Pi_f, \Pi_L\}$, arrival structure and waveguide interference. The shared object is the waveguide, not the Doppler.*

Mode count: in a Pekeris waveguide $f_m = (2m - 1)c/(4D\sqrt{1 - (c/c_b)^2})$, so with $\Pi_f = f/f_1$, $N_{\text{modes}} = \lfloor (\Pi_f + 1)/2 \rfloor$. In the ionosphere a hop mode propagates only *below* $f_{\text{MUF}} = f_{\text{crit}} \sec \varphi_0$, so a literal f/f_{cut} is not the same quantity; we propose each pack define Π_f so that N_{modes} is monotone in it (Check 4).

4 Generators

Sonar, image-source Pekeris [P]. Depth D , speed c , over half-space (c_b, ρ_b) . Image depths $z^{(n,\sigma)} = 2nD + \sigma z_s$, $n \in \mathbb{Z}$, $\sigma = \pm 1$; reflection counts $(n_s, n_b) = (|n|, |n|)$ for $\sigma = +$; $(n-1, n)$ for $\sigma = -$, $n \geq 1$; $(|n|+1, |n|)$ for $\sigma = -$, $n \leq 0$. Per image, $R = \sqrt{r^2 + (z_r - z^{(n,\sigma)})^2}$, $\tau = R/c$, $\theta = \arctan(|z_r - z^{(n,\sigma)}|/r)$ and

$$a = \frac{(-1)^{n_s} R_b(\theta)^{n_b}}{R}, \quad R_b(\theta) = \frac{\rho_b c_b \sin \theta - \rho c \sin \theta_b}{\rho_b c_b \sin \theta + \rho c \sin \theta_b}, \quad \frac{\cos \theta}{c} = \frac{\cos \theta_b}{c_b}, \quad (4)$$

with $|R_b| = 1$ and a frequency-independent phase below $\theta_c = \arccos(c/c_b)$, applied to the analytic source: $y = \Re \sum_k a_k s_a(t - \tau_k) + n$. Exact for the ideal waveguide; with $R_b(\theta)$ it is the ray approximation to the Pekeris field, omitting head waves and near-cutoff modal corrections, and it has no refraction [8, 12]. **HF, discrete-mode skywave [P].** Chapman layer $N(h) = N_m \exp\{\frac{1}{2}[1 - \zeta - e^{-\zeta}]\}$, $f_p \approx 8.98\sqrt{N}$; reflection where $f \cos \varphi_0 = f_p(h)$; group delay of an n -hop mode from the virtual height $h'(f_v) = \int_0^{h_r} dh / \sqrt{1 - f_p^2/f_v^2}$; $\tau_n = \frac{2n}{c} \sqrt{(R_g/2n)^2 + h'^2}$; O/X split as two reflection conditions; slow multiplicative fading per mode [4]. **Shared:** no code, different physics, one approximation family, discrete paths with per-path delay and gain, which is the anchor assumption; the independence statement must say so [G]. **Waveguide invariant [H]:** $\beta = -\Delta S_p / \Delta S_g$ with $S_p = k/\omega$, $S_g = dk/d\omega$ [5]; for any two paths with delay difference $\Delta\tau(R)$, $\beta_{\text{path}} = -(\text{d} \ln \Delta\tau / \text{d} \ln R)^{-1}$ reduces to it. H4 asks whether β_{path} is approximately invariant across HF hop-mode pairs and ground ranges.

5 Identifiability

Generative model $y_d = \mathcal{M}_d[\mathcal{F}_d(\theta_{\text{sh}}, \theta_{\text{pr}}; c_d)] + \varepsilon_d$; a change to $\mathcal{M}_d$ is lossy processing, to c_d a physical intervention, neither a “convention” [G]. Blind deconvolution $y = h * s$ admits $(\kappa h, s/\kappa)$, a common delay, and without structure $(h * g, g^{-1} * s)$; identifiability needs subspace or sparsity constraints [10]. A learned estimator hides an ambiguity, it does not remove one, so every estimator lists **assumptions, identifiable, ambiguous** [G]. Cepstrum: for two arrivals with $r = a_1/a_0$, $|r| < 1$, $\Delta = \tau_1 - \tau_0$,

$$\log(1 + re^{-j2\pi f \Delta}) = \sum_{n \geq 1} \frac{(-1)^{n+1} r^n}{n} e^{-j2\pi f n \Delta}, \quad (5)$$

impulses at $n\Delta$; it needs $|r| < 1$ (ordering is lost in the magnitude cepstrum), $\Delta > 1/B$, a smooth source spectrum, and one source, since $S_1 H_1 + S_2 H_2$ does not factor. Artificial time reversal uses near-linear modal phase and known geometry, leaving delay and scale unknown [13, 14]; bilinear models for sources of opportunity use a source subspace [16]. Multi-view identifiability [6, 17] requires views whose nuisance varies *independently*; a deterministic re-representation $\tilde{x} = \phi(x)$ is matched by $e \circ \phi^{-1}$ for any encoder e and isolates nothing. Hence L2a (consistency) is separate from L2b (content isolation), and only L2b is an alignment basis.

6 Transfer as a measured quantity

Loss at the independent unit u (scenes, sessions, instruments; never windows): $L_a(N) = \frac{1}{|U|} \sum_u \ell(\hat{q}_{a,u}, q_u)$ with $\ell = |\hat{\sigma}_\tau - \sigma_\tau|B$ in resolution cells; effect $\delta(N) = L_{\text{base}}(N) - L_{\text{cand}}(N)$ with a bootstrap CI over units and seeds [G]. H1: on O_q , δ exceeds a preregistered $\delta_{\min}$ with CI excluding zero at two or more budgets. H2: δ decreases with d_q ; a gain at large d_q is a finding, not a failed gate. H3: regime conditioning beats unconditioned arms under interventions separating regime from domain identity. H5: coverage survives shift or applicability flags the cells where it does not. Ladder: classical; normalized features + head; scratch; target SSL; pooled; dense shared/private; routed (optional), each compared to every arm below it [3, 7, 15]. $\delta_{\min}$ is set from the from-scratch arm’s seed variability and frozen before the test partition is read. A crash is not negative transfer; nonsignificance is not no effect; one simulator relabelled is not two domains. Fusion, two rules: shared-root results are one piece of evidence and the wider is kept; unknown dependence withholds numerical combination.

The nine checks

1. $\Pi_s = TB|\alpha - 1|$ as the delay–Doppler vs. delay–scale criterion; threshold of order 0.1–1? Does delay–scale lose modal structure at your ranges?
2. Is a local (non-WSSUS) scattering function needed as a first-class kind?
3. **Critical.** RMS delay spread and resolvable-arrival count as the first two descriptors; if not, which functional of P or C_H ?
4. Propagating-mode count as the common “modes above cutoff” coordinate across ocean and ionosphere?
5. Does medium variability need its own coordinate? Programme ranges for the groups.
6. **Critical.** Image-source Pekeris as the first generator, with your refraction-capable code second; $R_b(\theta)$ beyond critical as a constant phase on the analytic signal?
7. Should β_{path} be expected to be invariant in HF multimode returns; what would you look for first?
8. With overlapped sources, what breaks the cepstrum: delay collisions, source non-stationarity, or association? A source subspace that makes it identifiable? Below what Π_τ is sparsity unusable?

9. Does the per-descriptor pilot design lose anything from the original conversation? Descriptor-matched pairing across simulators as the primary basis, or pairs from a shared physical scene?

References

- [1] Richard A. Altes. Some invariance properties of the wide-band ambiguity function. *Journal of the Acoustical Society of America*, 53(4):1154–1160, 1973.
- [2] Philip A. Bello. Characterization of randomly time-variant linear channels. *IEEE Transactions on Communications Systems*, 11(4):360–393, 1963.
- [3] Chu et al. Do physics foundation models learn generalizable physics? *arXiv:2605.29283*, 2026.
- [4] Kenneth Davies. *Ionospheric Radio*. Peter Peregrinus (IET), 1990.
- [5] Gerald L. D’Spain and William A. Kuperman. Application of waveguide invariants to analysis of spectrograms from shallow water environments that vary in range and azimuth. *Journal of the Acoustical Society of America*, 106(5):2454–2468, 1999.
- [6] Luigi Gresele, Paul K. Rubenstein, Arash Mehrjou, Francesco Locatello, and Bernhard Schölkopf. The incomplete rosetta stone problem: Identifiability results for multi-view nonlinear ICA. In *Uncertainty in Artificial Intelligence (UAI)*, 2020. arXiv:1905.06642.
- [7] Ishaan Gulrajani and David Lopez-Paz. In search of lost domain generalization. In *ICLR*, 2021. arXiv:2007.01434.
- [8] Finn B. Jensen, William A. Kuperman, Michael B. Porter, and Henrik Schmidt. *Computational Ocean Acoustics*. Springer, 2nd edition, 2011.
- [9] Edward J. Kelly and Richard P. Wishner. Matched-filter theory for high-velocity, accelerating targets. *IEEE Transactions on Military Electronics*, 9(1):56–69, 1965.
- [10] Yanjun Li, Kiryung Lee, and Yoram Bresler. Identifiability and stability in blind deconvolution under minimal assumptions. *IEEE Transactions on Information Theory*, 63(7):4619–4633, 2017. arXiv:1507.01308.
- [11] Gerald Matz, Helmut Bölcskei, and Franz Hlawatsch. Time-frequency foundations of communications: Concepts and tools. *IEEE Signal Processing Magazine*, 30(6):87–96, 2013. arXiv:1307.4790.
- [12] Chaim L. Pekeris. Theory of propagation of explosive sound in shallow water. In *Propagation of Sound in the Ocean*, volume 27 of *Geological Society of America Memoirs*, pages 1–116. Geological Society of America, 1948.
- [13] Karim G. Sabra and David R. Dowling. Blind deconvolution in ocean waveguides using artificial time reversal. *Journal of the Acoustical Society of America*, 116(1):262–271, 2004.
- [14] Karim G. Sabra, H. C. Song, and David R. Dowling. Ray-based blind deconvolution in ocean sound channels. *Journal of the Acoustical Society of America*, 127(2):EL42–EL47, 2010.
- [15] Sharma and Sharma. Eradicating negative transfer in multi-physics foundation models via sparse MoE routing. *arXiv:2605.15179*, 2026.
- [16] Yuxuan Tian, Kiryung Lee, Justin Romberg, Nicholas C. Durofchalk, and Karim G. Sabra. Blind deconvolution of sources of opportunity in ocean waveguides using bilinear channel models. *Journal of the Acoustical Society of America*, 148(4):2267–2279, 2020.
- [17] Julius von Kügelgen, Yash Sharma, Luigi Gresele, Wieland Brendel, Bernhard Schölkopf, Michel Besserve, and Francesco Locatello. Self-supervised learning with data augmentations provably isolates content from style. In *Advances in Neural Information Processing Systems 34*, 2021. arXiv:2106.04619.